



Spatial Context and Nonlinear Attribution for Peak Learning in Mass Spectrometry Imaging

Zayd SULIMAN, Student No. 2653934

School of Computer Science and Applied Mathematics, Faculty of Science, University of the Witwatersrand

Supervisors: Hairong Bau and Richard Klein 2026

Abstract—Mass spectrometry imaging measures a high-dimensional mass spectrum at each tissue location, making reliable feature reduction necessary before downstream analysis. Existing variational-autoencoder peak learning is computationally accessible but spatially blind, whereas spatial convolutional methods use local structure at greater architectural complexity. This study separates two possible sources of improvement: adding local neighbourhood context to a lightweight variational autoencoder and replacing weight-based peak extraction with nonlinear attribution. Centre-only and eight-neighbour contextual models were trained under matched conditions on eight glioblastoma and eight colorectal-adenocarcinoma sections. A label-free two-component Gaussian mixture model was fitted to frozen latent means, and its posterior was attributed to mass-to-charge bins using Integrated Gradients. At matched peak counts, nonlinear attribution outperformed legacy msiPL on every section: mean mSCF1 increased from 0.3643 to 0.4562 on glioblastoma and from 0.4024 to 0.5595 on colorectal adenocarcinoma. Uniform neighbourhood context was not universally beneficial; it had no consistent glioblastoma advantage, but improved mSCF1 over the centre-only control on all eight colorectal sections. Corrected attention improved mSCF1 over uniform averaging across three training seeds on one development section, but did not improve deletion faithfulness and used more memory. The principal result is therefore a robust gain from nonlinear, label-free attribution, with spatial context providing a conditional rather than universal benefit.

Index Terms—mass spectrometry imaging, peak learning, variational autoencoder, spatial context, Integrated Gradients, explainable artificial intelligence

I. INTRODUCTION

Mass spectrometry imaging (MSI) maps molecular measurements directly to tissue coordinates, enabling biochemical structure to be studied without preselecting one target molecule [Caprioli et al. 1997; Schwamborn and Caprioli 2010]. Each pixel nevertheless contains a spectrum with hundreds to tens of thousands of mass-to-charge (m/z) bins. Many bins are redundant, noisy or unrelated to the biological structure of interest, so selecting a compact set of informative peaks is an important preprocessing step.

Conventional peak picking uses rules such as intensity and signal-to-noise thresholds. These rules are useful but can be

sensitive to user choices and may discard weak yet spatially coherent ions. Peak learning instead trains a model on spectra and derives a ranking from the learned representation. The msiPL approach uses a one-dimensional variational autoencoder (VAE), but processes each pixel independently [Abdelmoula et al. 2021]. S3PL explicitly consumes a spatial patch [Weigand et al. 2026], although its three-dimensional convolutional architecture and released implementation differ substantially from msiPL. This leaves two questions entangled: whether local context improves the learned representation, and whether a better explanation of a nonlinear representation improves the resulting peak list.

This report develops Spatial-msiPL, a controlled VAE study in which the central spectrum is compared with the same model supplied with a summary of its Moore neighbourhood. Rather than assuming that first-layer weights faithfully explain the model, a label-free Gaussian mixture model (GMM) is fitted in latent space and its posterior is propagated to input bins with Integrated Gradients [Sundararajan et al. 2017]. Expert masks are used only after ranking to quantify spatial peak quality.

The study answers three research questions:

- 1) Under matched training, how does eight-neighbour context affect reconstruction, latent tissue separation and matched-count peak quality relative to a centre-only VAE?
- 2) Does nonlinear GMM-targeted Integrated Gradients produce better peak rankings than legacy msiPL and simple first-layer L2 importance?
- 3) What peak-quality, explanation-faithfulness and computational trade-offs arise from learned attention, and are they stable across training seeds?

The aim is to determine separately when neighbourhood context and nonlinear attribution improve lightweight MSI peak learning. The principal contributions are: (i) a controlled centre-only/contextual comparison; (ii) a label-free GMM-targeted attribution pipeline; (iii) matched-count validation on eight GBM and eight CAC sections; and (iv) completeness, deletion-faithfulness, compute and training-seed analyses that bound the claims. The remainder of the report reviews related peak-learning and attribution methods, defines the controlled pipeline,

presents the experimental results, and discusses what they do and do not establish.

II. RELATED WORK

A. MSI feature reduction

MSI combines spectral and image structure, but its scale motivates feature reduction before visualisation, classification or biological interpretation. Traditional workflows include normalisation, smoothing, baseline correction and rule-based peak detection [Gibb and Strimmer 2012; Deininger et al. 2011]. Sparse and learning-based methods reduce manual threshold dependence, but a selected peak is useful for this study only when its ion image also reflects annotated tissue structure [Lieb et al. 2020; Winderbaum et al. 2015].

B. Representation and peak learning

VAEs learn a probabilistic low-dimensional representation by balancing reconstruction against regularisation of the latent distribution [Kingma and Welling 2014]. msiPL applies this idea to spectra and derives peaks through network-weight analysis [Abdelmoula et al. 2021]. Its small network is attractive computationally, but neighbouring pixels do not interact. S3PL introduces spatial self-supervision and patch-based convolutions, and also provides a correlation-based evaluation that converts ion-image agreement with tissue masks into mSCF1 [Weigand et al. 2026]. These methods motivate a lighter controlled test of spatial context inside an msiPL-style VAE.

C. Neural attribution

A first-layer weight norm measures only the model’s initial linear connection to each input. It cannot represent batch normalisation, nonlinear hidden transformations, latent geometry or the downstream decision surface. Integrated Gradients instead integrates the gradient of a chosen output along a path from a baseline input to the observed input and satisfies a completeness property under standard differentiability assumptions [Sundararajan et al. 2017]. The present work targets a label-free GMM posterior rather than a supervised class output. This lets attribution explain which m/z bins support structure learned without using expert tissue labels.

The project is therefore positioned as a controlled decomposition rather than a claim that a new architecture dominates every baseline: the same VAE and attribution procedure are used to isolate neighbourhood value, while matched peak budgets isolate ranking quality.

III. METHODOLOGY

A. System overview

Figure 1 separates training, label-free explanation and mask-based evaluation. No expert label is used to train the VAE, fit the GMM or produce the peak ranking.

TABLE I
DATASET CHARACTERISTICS USED BY THE EXPERIMENTS.

Collection	Sections	Bins	Classes	Coverage
GBM	8	85,062	2	54.62–82.95%
CAC	8	1,481	3	89.98–100%

B. Datasets and preprocessing

Two public MSI collections were used (Table I). The MassNet GBM data contain eight sections with normal/tumour annotations and a shared 85,062-bin axis [Abdelmoula et al. 2022]. Grid coverage ranges from 54.62% to 82.95%, so neighbourhoods are constructed from measured coordinates rather than by treating missing grid locations as spectra. The CAC collection contains eight sections, 1,481 bins and three mask classes. Seven grids are complete; 280TopL has 4,160 measured pixels in a 4,623-position grid (89.98% coverage). Coordinate-mask alignment and HDF5 round trips were audited before training.

Each measured spectrum s_i is total-ion-count (TIC) normalised,

$$x_i = \frac{s_i}{\sum_j s_{ij} + \epsilon}, \quad (1)$$

which reduces total-intensity scale differences while preserving relative bin composition. The expert masks are retained separately for evaluation.

C. Controlled VAE variants

The centre-only control receives x_i . The uniform contextual model computes the mean c_i of the valid spectra in the eight-position Moore neighbourhood and receives $[x_i, c_i]$. Missing neighbours contribute neither values nor weights. The corrected-attention variant learns neighbour weights and applies sqrt-bin input scaling before its scoring operation to avoid nearly uniform scores caused by the small TIC-normalised magnitudes. All variants use a 512-unit hidden layer and a five-dimensional Gaussian latent representation. The decoder reconstructs the central spectrum, so added context is useful only if it helps explain the same target.

For a sample x , the objective is

$$\mathcal{L} = \mathcal{L}_{\text{recon}}(x, \hat{x}) + \beta D_{\text{KL}}(q_\phi(z | x, c) \| \mathcal{N}(0, I)). \quad (2)$$

Matched comparisons use the same optimiser, schedule, split, epoch count and seed. Production VAE runs use 100 epochs. Explicit adjacent-latent penalties and Poisson augmentation were evaluated as development gates but were not promoted: the former forced smoothness while degrading reconstruction, and the latter did not reach its predeclared noisy-reconstruction improvement.

D. Label-free nonlinear attribution

After training, posterior means μ_i are frozen and a two-component GMM is fitted without tissue labels. For pixel i , the scalar attribution target is the posterior probability of its assigned component, $F_i = p(k_i | \mu_i)$. For input x and

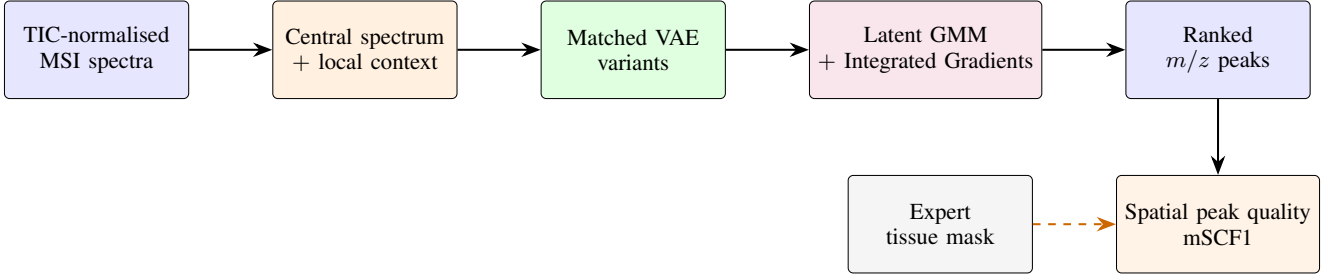


Fig. 1. Simplified Spatial-msiPL workflow. Each measured spectrum is TIC-normalised and supplied either alone or with a summary of its valid eight-position Moore neighbourhood. Matched VAE variants reconstruct the central spectrum. After training, a label-free two-component GMM is fitted to frozen latent means and its posterior is attributed to the original m/z bins using Integrated Gradients. The resulting ranking is evaluated at a matched peak budget. The dashed path emphasises that expert tissue masks are used only to calculate mSCF1 after peak selection, never to train the VAE, fit the GMM or generate the ranking.

section-mean baseline x' , Integrated Gradients for bin j is approximated by

$$\text{IG}_j(x) = (x_j - x'_j) \frac{1}{M} \sum_{m=1}^M \frac{\partial F(x' + \frac{m}{M}(x - x'))}{\partial x_j}. \quad (3)$$

Absolute attributions are aggregated by m/z across sampled pixels and GMM components. Component-specific rankings are combined deterministically. For a contextual model, contributions through the central and neighbourhood paths are mapped to the same spectral bins before aggregation.

Completeness checks whether summed attributions approximate $F(x) - F(x')$. Deletion faithfulness replaces highly ranked bins by baseline values and measures the reduction in assigned-component posterior; a larger drop indicates a more causally influential ranking. First-layer L2 importance is retained as a deliberately simple linear comparator.

E. Peak evaluation and statistical design

For every m/z bin, its ion image is correlated with each expert class mask using Pearson's r . At a threshold $t \in \{0.3, 0.4, 0.5, 0.6\}$, sufficiently correlated bins form a reference-positive set. Precision and recall of a selected peak list yield $F1_t$, and

$$\text{mSCF1} = \frac{F1_{0.3} + F1_{0.4} + F1_{0.5} + F1_{0.6}}{4}. \quad (4)$$

Within a section, compared methods select the same number of peaks. Results are paired by section and exact two-sided Wilcoxon signed-rank tests are reported where applicable. Section spread is not interpreted as patient-level independence. Broad GBM and CAC validation uses seed 1; a targeted GBM108-positive experiment repeats training with seeds 1–3 for centre-only, uniform-mean and corrected-attention variants.

IV. EXPERIMENTS AND RESULTS

A. Baseline reproduction

The tuned legacy msiPL reproduction obtained mean mSCF1 0.414 on GBM and 0.402 on CAC. These values lie inside the published aggregate intervals used for the reproduction audit. S3PL was executable on both collections, but its GBM mean remained below the reported paper result despite patch-size, normalisation, format and mask checks. The CAC S3PL

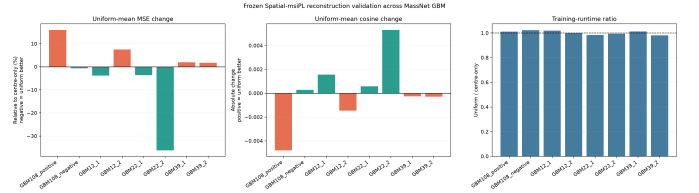


Fig. 2. Paired whole-GBM reconstruction comparison. Negative MSE change favours uniform context; the heterogeneous section effects produce no consistent aggregate advantage.

checkpoints were therefore re-evaluated at the same section-specific peak counts as the VAE methods; no retraining was needed. Matched S3PL averaged 0.5915 mSCF1.

B. Does uniform context improve the VAE?

On GBM, reconstruction MSE favoured uniform context in four sections and the centre-only model in four (exact Wilcoxon $p = 0.945$). Mean relative MSE change was -2.14% , but the median was $+0.52\%$ and one section strongly influenced the mean. Uniform context used 130,751,056 parameters and approximately 2.90 GiB peak GPU memory, compared with 87,199,312 parameters and 2.05 GiB for the centre-only model. Figure 2 shows that local context did not provide a consistent reconstruction advantage.

The peak-selection effect was also collection-dependent (Figure 3). On GBM, uniform-context IG averaged 0.4562 mSCF1 versus 0.4665 for centre-only IG, winning five of eight sections but with mean change -0.0103 . On CAC, it averaged 0.5595 versus 0.5285, won all eight sections and had a mean paired gain of 0.0310 (exact $p = 0.0078125$). Thus context changed the nonlinear ranking beneficially on CAC without a corresponding universal gain in reconstruction or GMM–mask agreement.

C. Does nonlinear attribution improve peak ranking?

Figure 4 summarises the primary matched-count result. On GBM, centre-only IG and uniform-context IG averaged 0.4665 and 0.4562 mSCF1, respectively, versus 0.3643 for legacy msiPL. At least one IG variant beat legacy msiPL in every section; uniform-context IG alone did so in all eight ($p = 0.0078125$). On CAC, centre-only and uniform-context IG

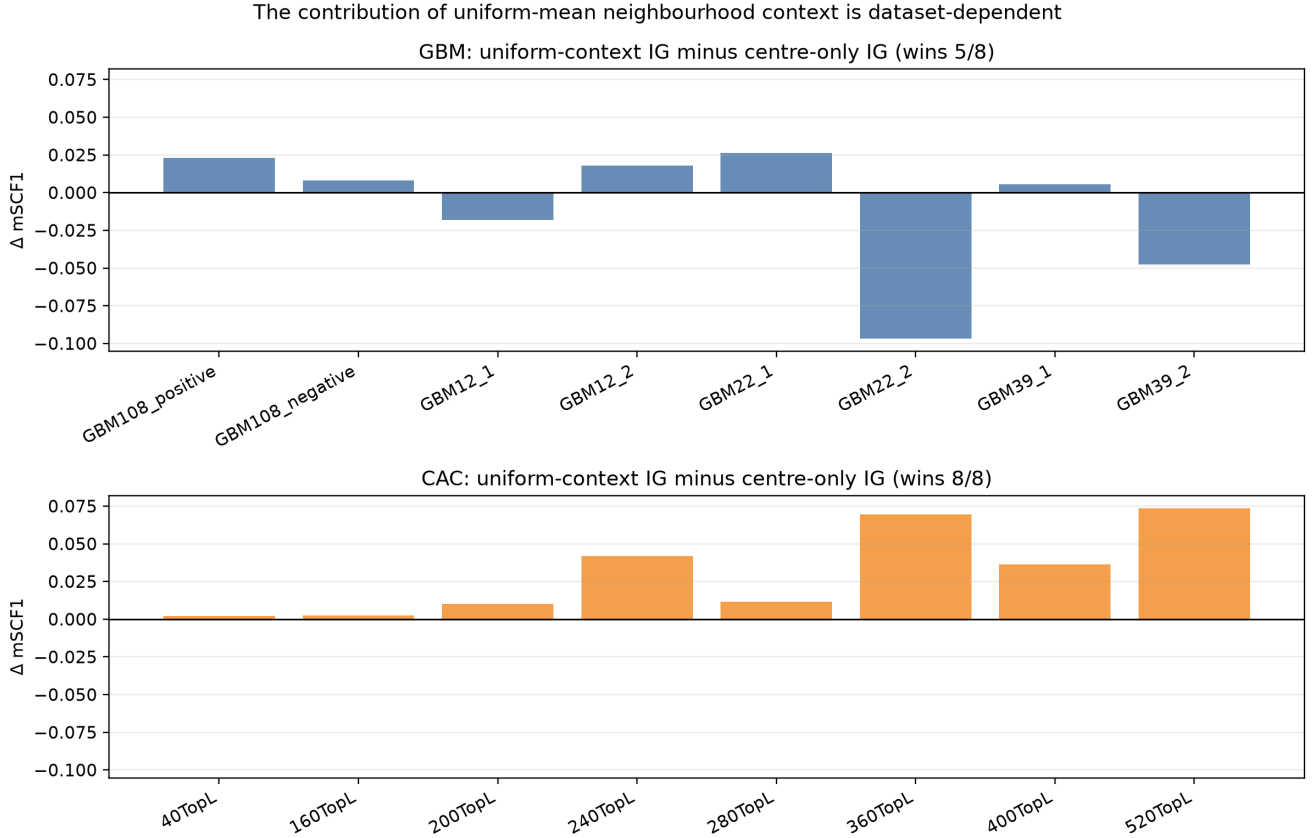


Fig. 3. Section-wise change in mSCF1 from adding uniform neighbourhood context to the otherwise matched VAE and GMM-targeted IG procedure. Context is mixed on GBM but positive on every CAC section.

averaged 0.5285 and 0.5595 versus 0.4024 for legacy *msiPL*; uniform-context IG again won all eight paired sections ($p = 0.0078125$).

First-layer L2 was generally much weaker than nonlinear IG (Figure 5). On the initial GBM108-positive test at 530 peaks, uniform-context IG achieved 0.528 mSCF1, legacy *msiPL* 0.404 and first-layer L2 0.068. IG passed the numerical completeness checks, and deleting IG-ranked features caused a larger GMM-posterior drop than deleting L2-ranked or random features. These results support the interpretation that gradients through the complete nonlinear encoder and latent decision surface contain information that is absent from the first layer alone.

S3PL remained stronger on matched CAC evaluation: 0.5915 versus 0.5595 for uniform-context IG, winning six of eight sections (exact $p = 0.0390625$). This comparison establishes peak quality, but not equivalent computational efficiency: S3PL used its released 10-epoch protocol whereas the VAE validation used 100 epochs.

D. Learned attention and training-seed stability

Corrected attention, uniform mean and centre-only models were independently trained with seeds 1–3 on the frozen GBM108-positive development section. Mean mSCF1 was 0.5120 ± 0.0179 , 0.5250 ± 0.0132 and 0.5454 ± 0.0188 (sample standard deviation), respectively (Figure 6). Attention beat uniform mean in all three paired seeds, with gains +0.0347,

+0.0120 and +0.0144 (mean +0.0204). Mean GMM balanced accuracy was 0.8745, 0.8735 and 0.8822.

The attention result did not pass the combined predeclared gate. Its mean deletion-faithfulness score was 0.0045 below uniform mean and it won that metric in only one seed. Peak allocated GPU memory also increased from about 2.90 GiB for uniform mean to 3.20 GiB for attention. The result is therefore a stable peak-quality improvement on one section, not evidence of a uniformly superior model.

V. DISCUSSION

The clearest result is not that spatial input always improves a VAE. It is that the way a learned representation is explained matters strongly for peak selection. Both centre-only and contextual GMM-targeted IG exceeded legacy *msiPL* across the complete GBM and CAC evaluations, while first-layer L2 often failed. This is consistent with the methodological difference: L2 observes one linear transformation, whereas IG follows the full nonlinear path to a label-free latent-space decision.

RQ1 has a conditional answer. Uniform context did not consistently improve GBM reconstruction or mSCF1, but improved CAC mSCF1 in every paired section. A neighbour mean can reduce local noise when adjacent spectra share relevant structure, but can also blur boundaries or combine heterogeneous tissue. The different spectral resolution, tissue classes, coverage and acquisition properties of GBM and CAC

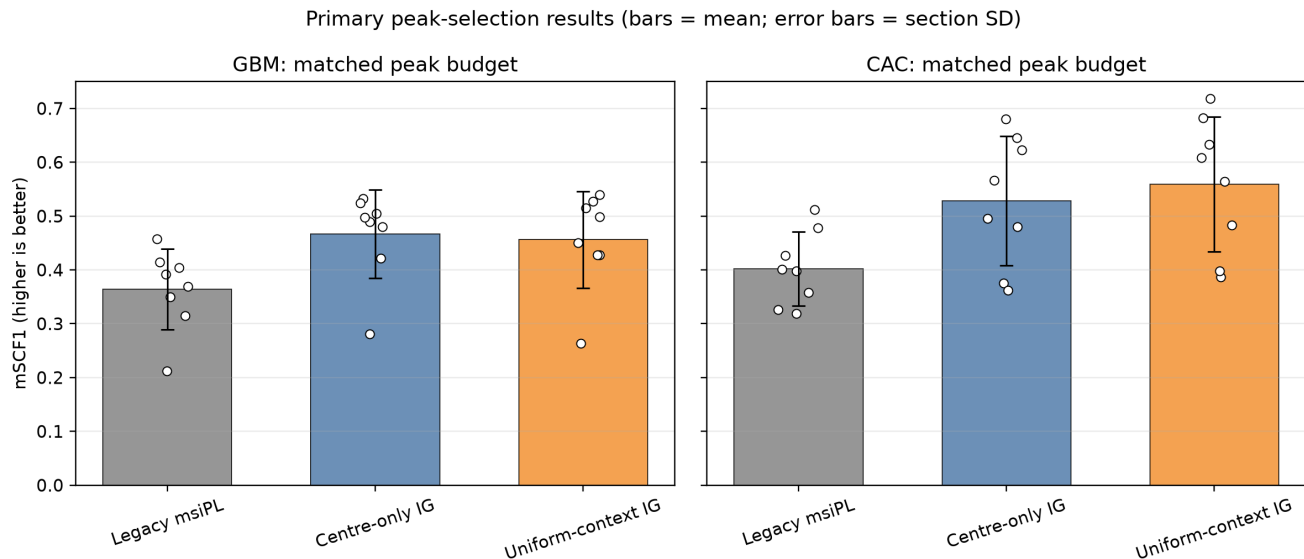


Fig. 4. Matched-count peak-selection quality across the two collections. Markers represent tissue sections; bars summarise method-level means. The consistent comparison is within collection and section, not between the two different datasets.

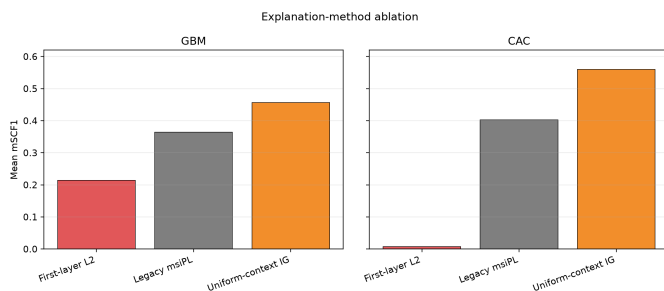


Fig. 5. Explanation-method ablation at matched peak counts. The simple first-layer L2 shortcut is not an adequate substitute for GMM-targeted nonlinear attribution.

plausibly change that balance. The experiments show the boundary of the effect; they do not identify one biological cause.

RQ2 is supported more broadly. Nonlinear IG improved matched-count peak quality over legacy msiPL across both collections. Completeness and deletion tests add important evidence beyond mSCF1: the former checks numerical consistency with the explained output, while the latter asks whether removing highly ranked bins actually changes that output. Nevertheless, a high mask-correlation score does not validate a peak as a biomarker. Independent molecular identification and biological validation remain outside this study.

RQ3 shows a trade-off. Corrected attention gave a small but seed-consistent mSCF1 improvement on the development section, yet did not improve deletion faithfulness and consumed more memory. It should therefore remain an ablation until tested across independent sections. Uniform averaging remains the more defensible broad-validation model because it is parameter-free and was frozen before whole-dataset evaluation.

S3PL is the strongest executable CAC baseline in peak quality. The present method should not be described as both

more accurate and cheaper than S3PL: its matched CAC mSCF1 is lower, and the 10- versus 100-epoch protocols make the recorded runtimes non-equivalent. The value of Spatial-msiPL is instead the controlled evidence about context and attribution within a VAE framework.

Limitations include seed-1 section-wide validation, three-seed testing on only one development section, sections that are not independent patients, dataset-specific peak budgets, a GBM S3PL reproduction gap, and variable cluster hardware availability. Future work should repeat training across patients and sections, constrain attention more efficiently, investigate automatic peak-budget selection, and validate selected ions biologically.

VI. CONCLUSION

This study separated neighbourhood input from nonlinear peak attribution in a lightweight MSI VAE. For RQ1, uniform eight-neighbour context was not a universal improvement: GBM reconstruction and mSCF1 were mixed, while CAC mSCF1 improved in all eight sections. For RQ2, label-free GMM-targeted Integrated Gradients was the robust contribution, improving matched-count peak quality over legacy msiPL throughout both collections and substantially outperforming simple first-layer L2 importance. For RQ3, corrected attention improved mSCF1 over uniform averaging across three training seeds on one development section, but its faithfulness and memory trade-offs prevented its promotion as the final general method. S3PL remained stronger on matched CAC peak quality. The resulting conclusion is deliberately bounded: nonlinear attribution makes the VAE representation more useful for peak ranking, whereas the value of spatial context depends on the dataset and aggregation mechanism.

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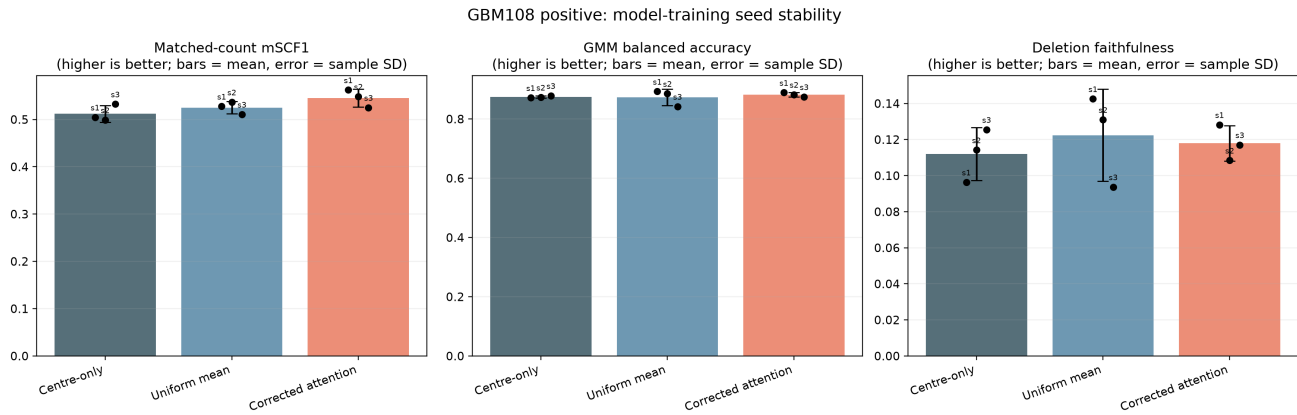


Fig. 6. Targeted three-seed stability on GBM108-positive. Corrected attention improves matched-count mSCF1 over uniform averaging in every seed, but the experiment is restricted to one development section.

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